

Java

What is Java ?

- Java is a high-level, object-oriented programming language.
- It was developed by Sun Microsystems in 1995, and today it's maintained by Oracle.
- It's widely used in web development, mobile apps (Android), enterprise software, and even banking systems.

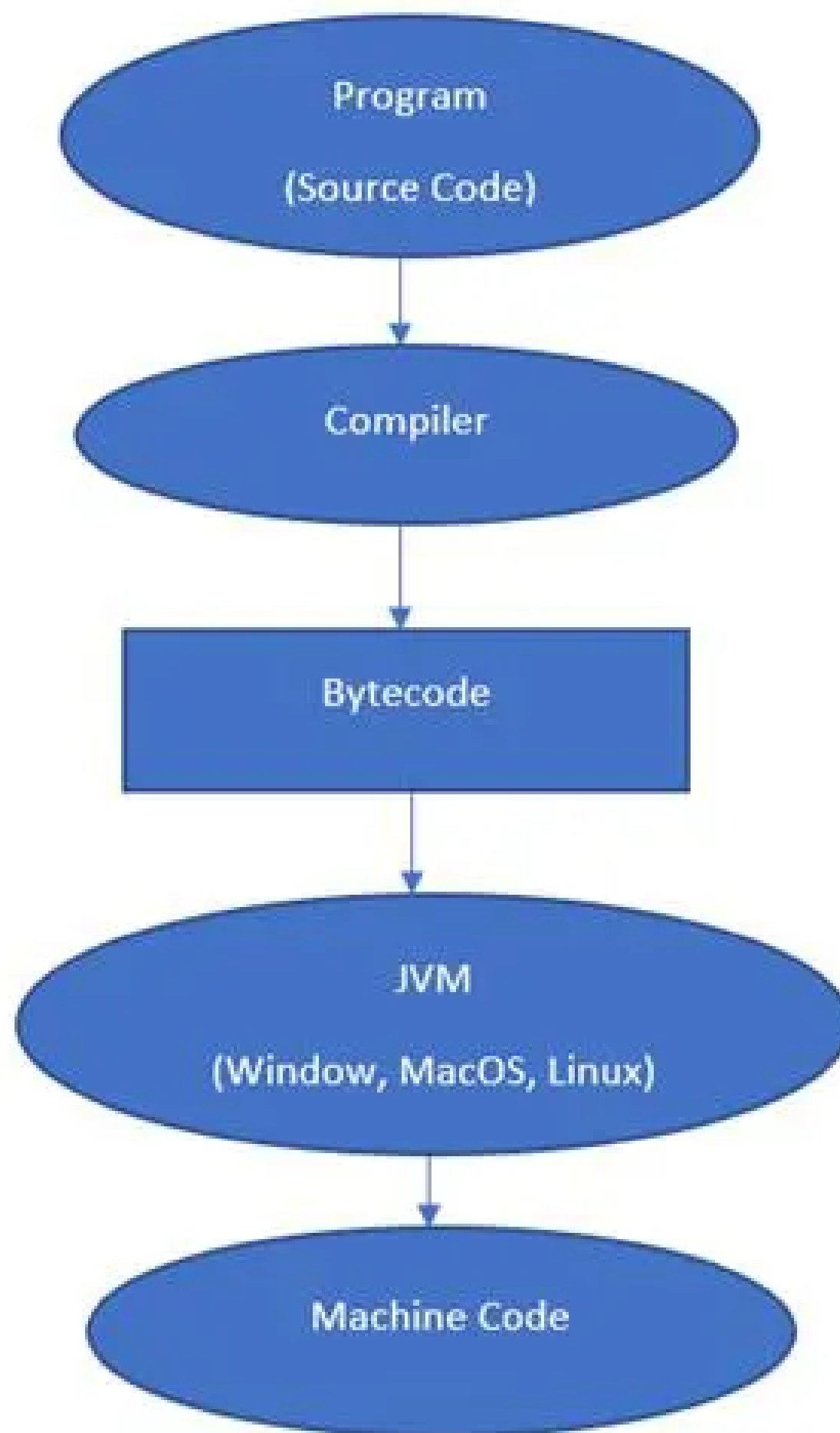
Features of java

Java is still in demand today because of these powerful features:

- **Object-Oriented** – Based on real-world modeling.
- **Platform Independent** – Write Once, Run Anywhere (WORA).
- **Robust & Secure** – Used in banking, finance, enterprise apps.
- **Huge Community Support** – Thousands of libraries, frameworks.
- **Scalable** – Used by companies like Amazon, Netflix, Google.

Where Java is Used in Real Life

- **Android Apps** – Java is the official language for Android.
- **Banking & Finance** – For backend logic.
- **Web Applications** – With Spring Boot & other frameworks.
- **Game Development** – Minecraft was built using Java!
- **Data Engineering & Big Data Tools**



Installing JDK and IDE

Install JDK

windows [https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.exe%20\(shsha256\)](https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.exe%20(shsha256))

Installing VS code

windows <https://code.visualstudio.com/docs/?dv=win64user>

JDK and JRE

JRE – Java Runtime Environment

- **Purpose:** It is used to run Java programs.
- **Contains:**
 - Java Virtual Machine (JVM)
 - Core libraries
 - Supporting files
- **Use case:** If you just want to run Java apps (not develop), you only need the JRE.

JDK and JRE

JDK

- java Development Kit.
- Purpose: It is used to develop Java programs.
- Contains:
 - Everything in JRE (JVM + libraries)
 - javac (Java compiler)
 - Debugger, documentation tools, etc.
- Use case: If you want to write and run Java code — like you are doing now — you need the JDK.